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Redline Rotary's
Rotary Engine Design Project

Phase 1

https://youtu.be/27UCl4HXXh4
https://drive.google.com/file/d/1olsIIr02lB-4Txo-qj8AVUWLu-WaECbp/view?usp=sharing

Executive Summary

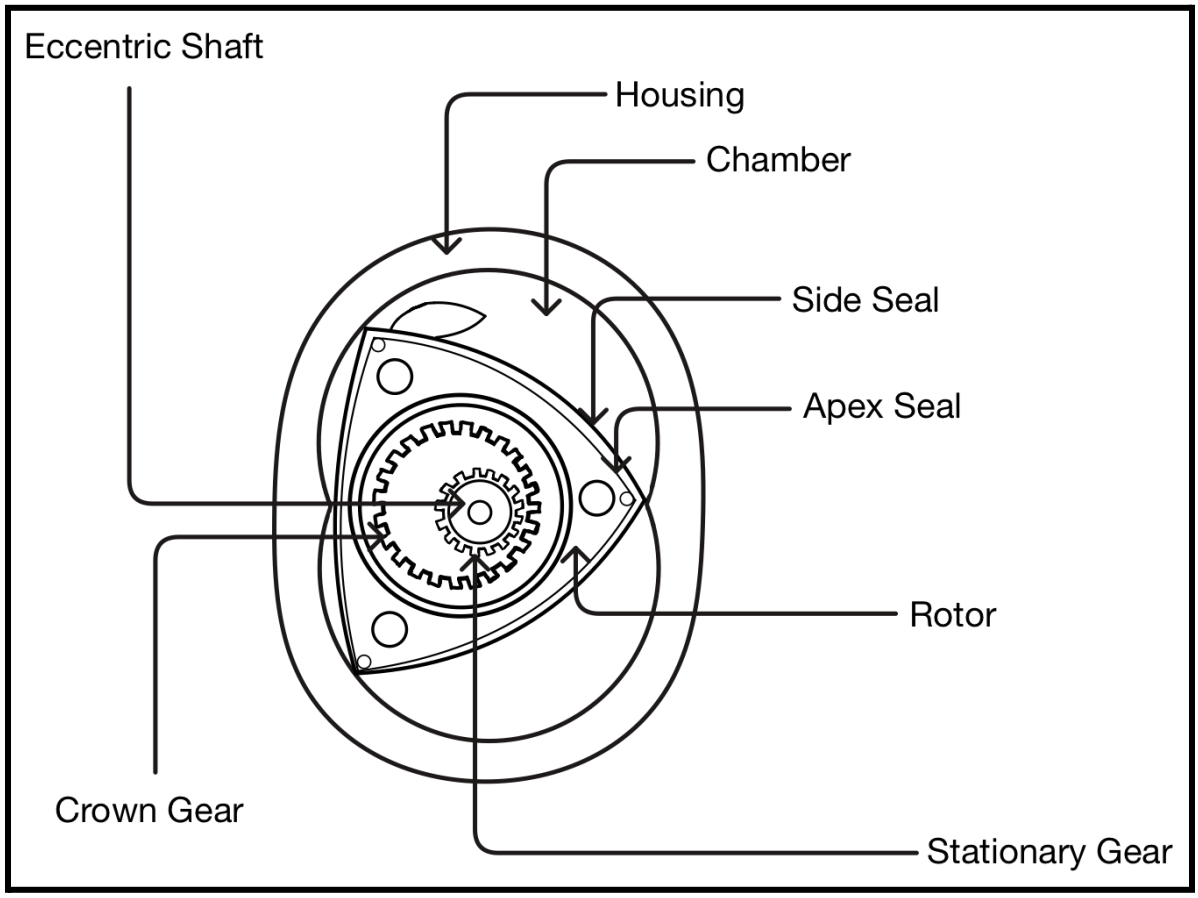
This report presents Phase 1 of the Redline Rotary Wankel Engine Project.

How the Wankel Rotary Engine works

Much like a conventional piston internal combustion engine, the purpose of the wankel rotary engine is to convert chemical energy from gasoline into mechanical energy. Also similarly, the engine does this by compressing and igniting air-fuel mixtures to harness the explosion's energy as torque. Where the wankel rotary engine differs from traditional ICE's is in its geometry; resulting in a unique set of advantages and disadvantages.

This engine is a reciprocating, four-stroke system that operates three combustion chambers per rotor assembly. That means that each unit of this engine intakes air-fuel mixture, compresses it, ignites it to harness energy, then forces out exhaust gases cyclically in three different instances concurrently. It accomplishes this using an epitrochoid-shaped housing with a curvilinear equilateral triangle shaped rotor inside that rotates on a 3:1 speed ratioed eccentric crankshaft. This system is then completed by sealing the combustion chamber, adding intake and exhaust ports, then typically a spark-plug ignition system and appropriate cooling and lubrication systems.

Key Components



Housing (aluminum with chrome plating)
Rotor (cast iron)
Eccentric shaft (machined steel)
Stationary gear (hardened cast iron)
Rotor bearing (aluminum)
Spark plug (iridium aluminum oxide)
Apex seal (and springs) (graphite)
Side seal (and springs) (steel)
Coolant/Oil O-rings (viton)

Subsystems

Fuel system - This engine requires a precise air-fuel mixture to achieve proper combustion. Fuel is mixed with air through a carburetor or fuel-injection system. A crankshaft angle sensor, exhaust gas oxygen content (lambda) sensor, and air pressure sensor may be used to optimize fuel quantity and mixture timing. Many wankel systems mix lubrication into air-fuel mixture much like a two-stroke engine.

Lubrication system - The rotor bearing is the main point of lubrication in a wankel rotary engine. The bearing is oiled by passages through the middle of the crankshaft that contain pressurized oil from the oil pump. Oil then circulates around the rotor bearing and through the inside of the rotor. Oil can also be mixed into the air-fuel mixture to lubricate external rotor seals like the apex and side seals.

Spark system - Very few compression ignition wankel rotary engines exist, meaning spark ignition is common practice due to consistency and engine knock prevention. Either crank angle sensor or magneto spark systems are implemented to ignite non-CI fuels.

Cooling system - Depending on application, water cooling may be needed to keep engine temperatures within operating range. Smaller UAV applications utilize fins for air cooling, while automotive applications use coolant passages, a water pump, and a radiator. An oil cooler is sometimes utilized to include cooling from the lubrication core of the system.

Failure Modes

Seal destruction:
- Detonation (usually due to overheating)
- Lubrication problems

Bearing destruction:
- Lubrication problems
- Revving too high

Overheating/thermal distortion:
- Poor cooling design
- Uneven heat rejection

Oil Metering Failure:
- Pump/injector malfunction

The main failure mode is the apex seals, which provides sealing between each chamber on the tips of each triangular rotor. This can be caused by wear, improper lubrication between the components, or excess heat, which eventually leads to compression loss and power drop. Another cause of apex seal failure is detonation, which is when the air-fuel mixture combusts prematurely; the pressure from the detonation can shatter the apex seal while it is not in position to take that kind of force. This failure is catastrophic to engine operation, with reassembly only possible if loose seal pieces do not gash housing and rotor components.

The most important components to minimize failure are the material used for the apex seals, the apex seal springs, and the fuel, lubrication, and cooling systems. The apex seal material used changes the wear resistance and longevity of the seals. The seal springs are important because

they make sure that the seals maintain consistent contact with the rotor, optimizing compression. All of the fuel, lubrication, and cooling systems are important components because if they are not operating properly, it could cause premature wear/overheating to the engine which would cause it to fail.

Critical design Parameters

Our design will be a scaled-down version of Mazda's 13b wankel rotary engine.

The gear ratio of stationary gear to rotor gear is 2:3 with a teeth count of 36:54 some other critical dimensions being a radius of 105 mm from rotor center to apex seal point, chamber/rotor width of 80 mm.
Eccentric shaft diameter (rotor bearing diameter -> clearance)
Engine displacement 1.3L
15mm crank offset (stroke)
Sealing apex and side seals (too large of seals -> interference damage, too small -> no compression)
Epitrochoidal housing and curvilinear equilateral triangle rotors with widths of 80mm
Crankshaft spins 3x as fast as the rotor, so one rotations of the rotor is 3 rotations of the crank

Applications

UAV - Single rotor, air cooled, small, high power to weight ratio, low vibration, fuel versatility
Automotive - 2 or 3 rotor, coolant and oil cooled, larger displacements
Hybrid Power Generation - engine can act as a generator for electric motors with high fuel efficiency, can be used as a range extender

Kinematics

The kinematic relationship between the rotor and the stationary gear can be described as

$$\omega_r = \omega_c\left(1 - \frac{N_s}{N_r}\right)$$

Where:

ω_r is the angular velocity of the rotor

ω_c is the angular velocity of the eccentric shaft

N_s is the number of teeth on the stationary gear

N_r is the number of teeth on the rotor gear

In most wankle applications, $\frac{N_s}{N_r} = \frac{2}{3}$

Sources

https://patents.google.com/patent/US5305721A/en